Course: Data Science and ML **Assignment:** *Data Science Portfolio Assignment* **Student ID number:** *BMT28011, BMT28004, BMT28048, BMT28052*	**Date:** *15/10/2026* **Professor:** *Hunter Yan*

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Analysing Air Quality and Its Link to Urban Factors

Data Science (DSAI501) Group Assignment 1, Option B

Group members and student IDs

Student IDs: BMT28011, BMT28004, BMT28048, BMT28052

Chosen problem statement: Option B, Analysing Air Quality and Its Link to Urban Factors

Date of submission: 15/10/2026

Executive Summary

We used a year of hourly readings from a roadside air quality monitor in an Italian city to find out when carbon monoxide (CO) pollution is high and which factors go with it. The data came from the UCI Air Quality dataset (Vito, 2008) and held 9,357 hourly records from March 2004 to April 2005. We cleaned the data, documented how we handled the gaps, explored daily and seasonal patterns for four pollutants, and trained four regression models to predict CO, with Linear Regression and Random Forest as our two main models.

The clearest finding is that the time matters far more than the weather. CO averaged 3.7 mg/m³ at 19:00 and only 0.7 mg/m³ at 05:00, and weekday CO was about 30% higher than weekend CO. Models using only time and weather explained about a third of the variation in CO (R^2 0.36 to 0.40, RMSE about 1.0 mg/m³). Adding five cheap chemical sensors lifted R^2 to 0.86 and cut the error to 0.49 mg/m³ with our best model, a tuned Random Forest.

Our main recommendation is that the city should target weekday rush hours and the winter months instead of whole days. The site never broke the EU 8-hour CO limit, yet it passed the stricter WHO guideline on 15 days and broke the EU hourly nitrogen dioxide limit on 87 days. Reporting only the CO legal limit would make this site look clean when it is not.

1. Introduction

Governments collect air quality data, and planners use it to decide on traffic policy, industrial zoning and emergency alerts. But raw sensor data is messy, and pollution changes by hour, by day, by season and with the weather.

Our aim was to turn those patterns into something a city could act on. We asked three questions. How much of the data is usable, and how should gaps be handled? Which times and conditions go with high pollution? How well can a regression model predict CO, and what would that mean for a real decision?

We chose CO as our target because it is measured by a certified reference analyser, which makes it the most trustworthy pollutant in the dataset. The dataset comes from a multi-sensor device placed at road level in a polluted area of an Italian city (De Vito et al., 2008). It has three kinds of columns. The first is reference analyser readings, marked (GT) for ground truth, for CO, benzene, nitrogen oxides (NOx) and nitrogen dioxide (NO2). The second is the responses of five low-cost metal-oxide sensors. The third is temperature (T), relative humidity (RH) and absolute humidity (AH). The data is published under a CC BY 4.0 licence and used here only for education.

2. Data Description and Preprocessing

We loaded the data with the ucimlrepo package, so we worked with exactly the published file. It has 9,357 rows, one for each hour from 10 March 2004 to 4 April 2005, and 15 columns. Two problems appeared straight away. Date and time were stored as separate text columns, so we combined them into one datetime index. Missing values were stored as the number -200 instead of blanks. If left alone, every average and every model would have been pulled towards -200, so we replaced all of them with NaN.

Missing value analysis

We asked how much is missing, where, when and for how long, because each answer changes the right fix. The share of missing hours fell into three groups (Figure 1). NMHC(GT) was 90.2% missing, because that analyser was effectively switched off after the first weeks. The CO, NOx and NO2 reference analysers were each about 18% missing (18.0%, 17.5% and 17.5%). Every sensor and weather column was missing the same 366 hours (3.9%), which shows that the whole device went offline at those moments.

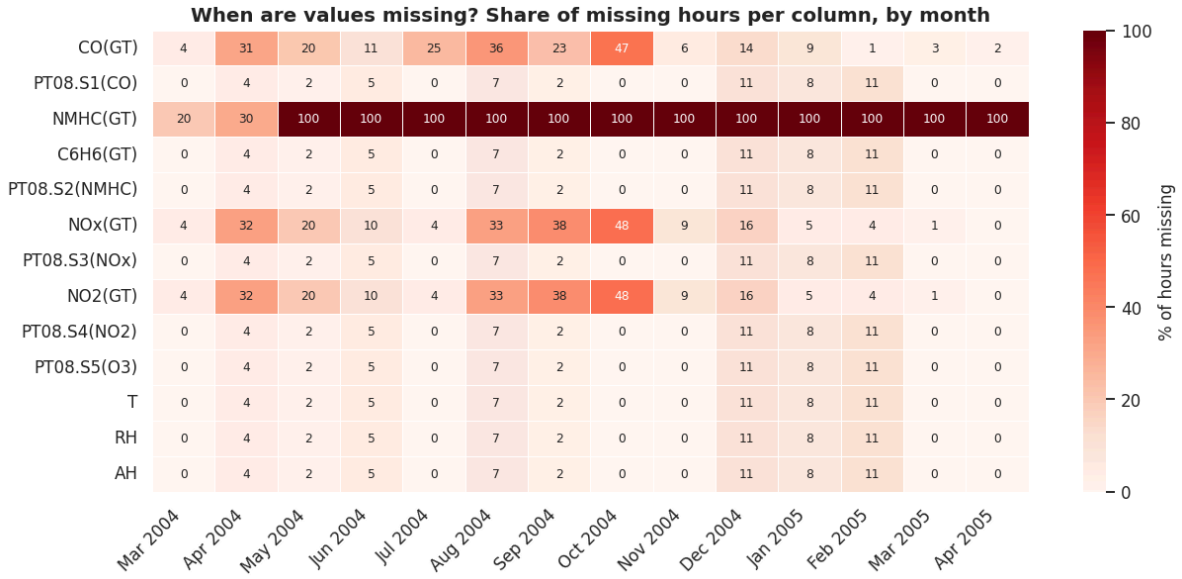


Figure 1. Share of hours missing for each column, by month. Gaps are clustered in time, not random.

The gaps were not random. NMHC(GT) almost completely stops from May 2004, and the CO, NOx and NO2 analysers lose between 20% and 48% of their hours in April, August, September and October 2004. For CO there were 187 gaps, and 86% lasted six hours or less. But those short gaps held only 11% of the missing hours. The rest sit in a few long outages, the longest lasting 173 hours, about a week.

Imputation strategy

Our strategy follows from those findings, and we applied a different rule to each group of columns.

- NMHC(GT) was dropped. With about 90% missing, there is nothing to fill it from, and any fill would be invented data.
- The target, CO(GT), was never imputed. Filling it would mean training the model on values we made up. Rows with no CO reading are left out of modelling only.
- NOx, NO2 and benzene were left as NaN for the charts, since averages simply skip them. We did not use them as model inputs, because a city without a CO analyser would not have those analysers either.
- Temperature, humidity and the five sensors were filled by linear interpolation in time, but only inside gaps of six hours or fewer. These variables change smoothly from hour to hour, so a straight line between neighbours is a fair estimate for a short gap. Over six hours a whole daily cycle could be missed, so longer gaps stay empty.

The six-hour limit comes from the gap length chart. The rule filled only 15 hours in each of these columns, and 351 hours stay missing. The distributions of temperature and humidity were unchanged by the filling, which is what we want: short holes closed, long ones not guessed.

Outliers and data types

We found no physically impossible values, such as negative concentrations or humidity outside 0 to 100%. The IQR rule flagged a few hundred high hours (215 for CO and 435 for NOx), but we kept them. They are the real rush-hour and winter episodes a city cares about, and removing them would remove what we are trying to study.

3. Exploratory Data Analysis

The year and the seasons

All four pollutants share one seasonal shape, as Figure 2 shows. They are lowest in July and August, rise sharply from September, peak between October and December, and ease through spring. Monthly mean CO rose from 1.28 mg/m³ in

August 2004 to 2.75 mg/m³ in December 2004, which is more than double. NOx showed the biggest swing, from weekly means below 60 ppb in late summer to above 600 ppb in November. Winter heating and cold, still air are likely reasons. The four pollutants also move together (CO and benzene correlate at 0.93, and CO and NOx at 0.78), which points to a shared source, mainly road traffic.

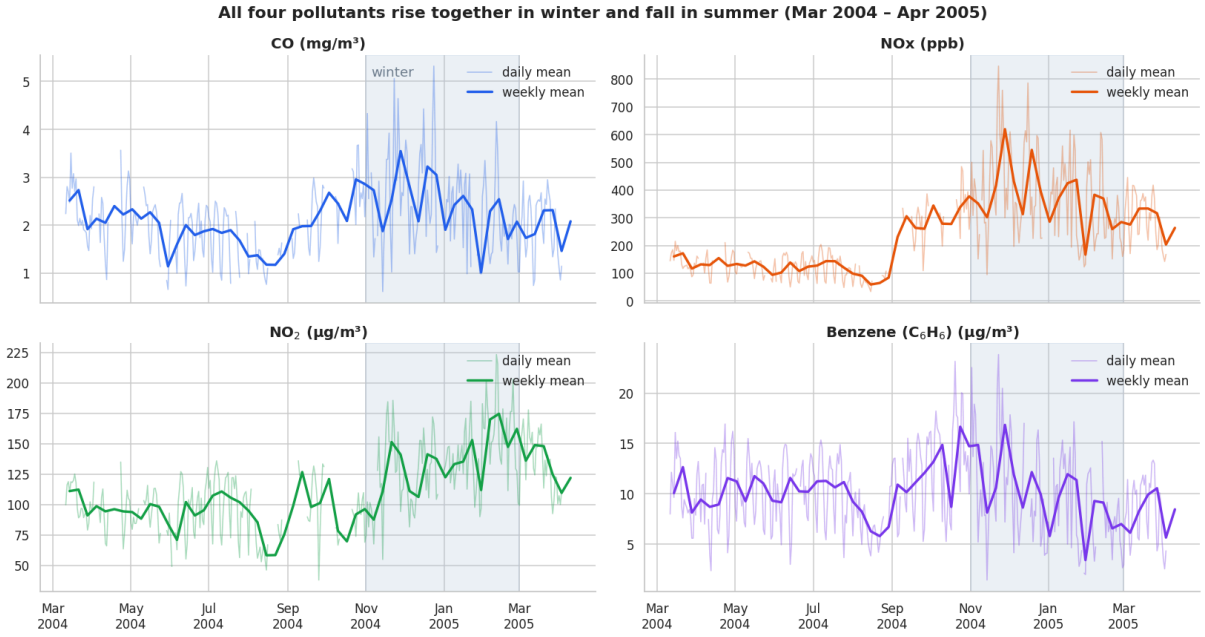


Figure 2. Daily and weekly mean of four pollutants, March 2004 to April 2005. The shaded band marks winter.

Hours of the day and weekends

The daily pattern is stronger than the seasonal one. Weekdays have two peaks, one around 08:00 to 09:00 and a larger one at 19:00, when average CO reaches 3.7 mg/m³, compared with 0.7 mg/m³ at 05:00. These are commuting hours. Weekends differ in shape and not just in level. The morning peak almost disappears, with weekend CO at 08:00 being 60% lower than on weekdays, and the evening peak is smaller and later. Between midnight and 04:00 the order flips, and weekends are 35% to 60% higher, which fits late-night social traffic.

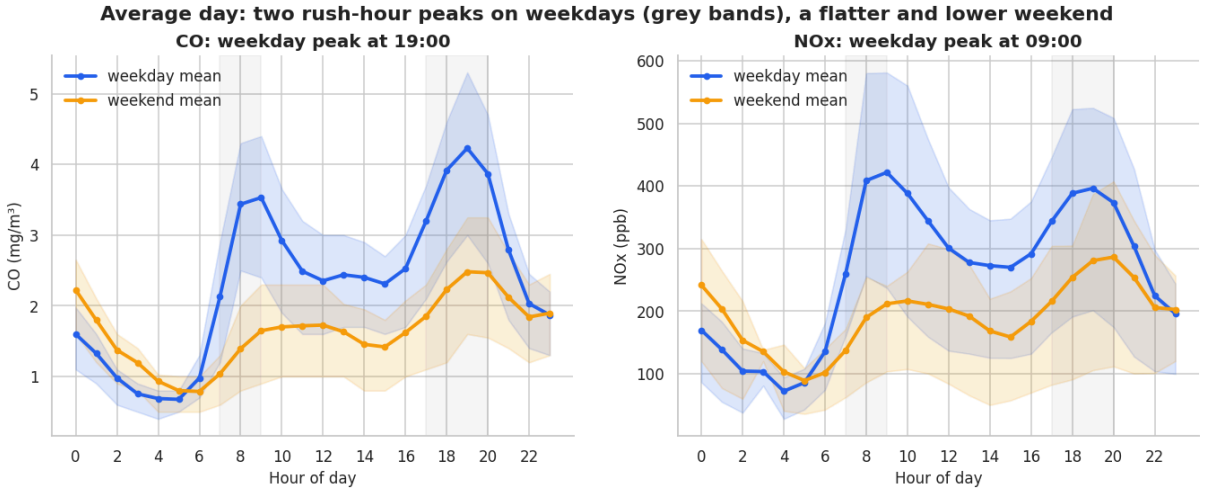


Figure 3. Average day for CO and NOx on weekdays and weekends. Shaded bands are the interquartile range.

We tested these patterns instead of only describing them. Daily mean CO was 2.36 mg/m³ on weekdays and 1.66 mg/m³ on weekends, a 30% drop (Mann-Whitney U test, $p < 0.001$). Hour of day and month both had large effects (Kruskal-Wallis tests, $p < 0.001$). We used non-parametric tests because pollution data is skewed.

Weather

Across the whole year, weather barely mattered. The Spearman correlations with CO were +0.07 for temperature, -0.01 for relative humidity and +0.05 for absolute humidity. The coldest hours were the cleanest, because they are mostly night hours with no traffic, so the time of day hides the weather effect. The dataset has no rainfall or particulate matter, so we could not answer the rainfall question in the brief directly and tested humidity and temperature instead.

Sensor drift

We also checked whether the cheap sensors can be trusted over time. The PT08.S1(CO) sensor stayed well correlated with the reference analyser each month (r between 0.84 and 0.95). But correlation is not accuracy. Calibrated on the first three months, it under-read by about 0.6 mg/m³ in November and December, roughly 20% of the winter average (Figure 4). A city that calibrated once and then trusted the sensors would under-report pollution when it matters most.

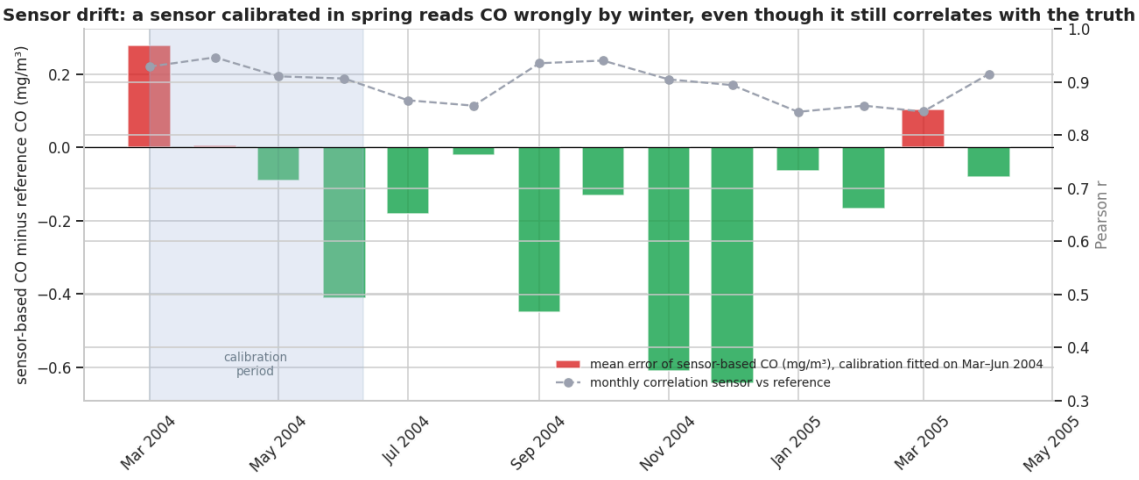


Figure 4. Monthly error of sensor-based CO after calibrating on March to June 2004.

Legal limits

Finally, we counted how often official limits were broken. The EU 8-hour CO limit of 10 mg/m³ (European Parliament and Council, 2008) was never exceeded. The WHO 24-hour CO guideline of 4 mg/m³ (World Health Organization, 2021) was exceeded for 166 hours spread over 15 days, in November, December and February. The EU hourly NO2 limit of 200 µg/m³ was exceeded in 386 hours on 87 days, which is 5.0% of observed hours, and February 2005 alone had 160 of them. The annual mean of benzene was 10.08 µg/m³, which is double the EU limit of 5 µg/m³.

4. Modelling Approach

Our target was CO(GT), and we compared two sets of inputs. Set A used only urban factors a planner could know in advance: hour of day, day of week, weekend, public holiday and rush-hour flags, season, and the three weather variables. This set answers the question of in which situations pollution will be high. Set B added the five cheap sensor readings, which tells us how much a low-cost sensor box adds. We left out the other pollutants, since a city without a CO analyser would not have those either, and past CO values, since that would turn the task into forecasting, which the brief does not require.

Each feature had a reason. Hour of day has two peaks, so a single sine wave cannot describe it, and we one-hot encoded hour, weekday and season for the linear models. We added a rush-hour flag (07:00 to 09:00 and 17:00 to 20:00, fixed in advance) and an Italian public holiday flag, because traffic on those weekdays behaves like a weekend. We also checked multicollinearity with the variance inflation factor. Temperature scored 12.4 and absolute humidity 8.5, because absolute humidity is calculated from temperature and humidity. We dropped it for the linear models, which brought the other scores down to 1.5.

We trained four models: Linear Regression, Ridge Regression, Random Forest and Gradient Boosting. Linear Regression and Random Forest are the required pair: the first is a simple baseline with high bias and low variance, and the second is an ensemble of trees with lower bias and higher variance. We added two simple baselines, the training mean and the average for each hour, so the scores have a reference.

The most important choice was how to split the data. Neighbouring hours of pollution are nearly identical, so a random split would put near-copies of the test hours into the training set. We split by time instead. The first 80% (5,886 hours, 10 March 2004 to 28 January 2005) trained the models, and the last 20% (1,472 hours, 28 January to 4 April 2005) tested them. We tuned the Random Forest with time-ordered cross-validation inside the training period. The scaler, encoder and model sit in one scikit-learn pipeline fitted on training rows only, so nothing leaks from the test set. Of the 9,357 hours, 7,358 were usable.

5. Results and Discussion

We evaluated all models on the held-out period using RMSE, which is in mg/m³ and punishes large misses, and R^2, which is the share of hour-to-hour variation the model explains. MAE is the typical absolute miss. Table 1 and Figure 5 show the results.

Model	Inputs	Test R^2	Test RMSE	Test MAE
Baseline: training mean	None	-0.031	1.319	1.068
Baseline: hour-of-day average	Hour only	0.293	1.093	0.838
Linear Regression	A: urban factors	0.359	1.040	0.817
Ridge Regression	A: urban factors	0.359	1.040	0.817
Random Forest	A: urban factors	0.370	1.031	0.775
Gradient Boosting	A: urban factors	0.395	1.010	0.759
Linear Regression	B: plus cheap sensors	0.789	0.597	0.470
Ridge Regression	B: plus cheap sensors	0.803	0.577	0.453
Random Forest	B: plus cheap sensors	0.854	0.497	0.337
Gradient Boosting	B: plus cheap sensors	0.849	0.506	0.352
Random Forest (tuned)	B: plus cheap sensors	0.858	0.490	0.334

Table 1. Test results on the held-out period (28 January to 4 April 2005). RMSE and MAE are in mg/m³.

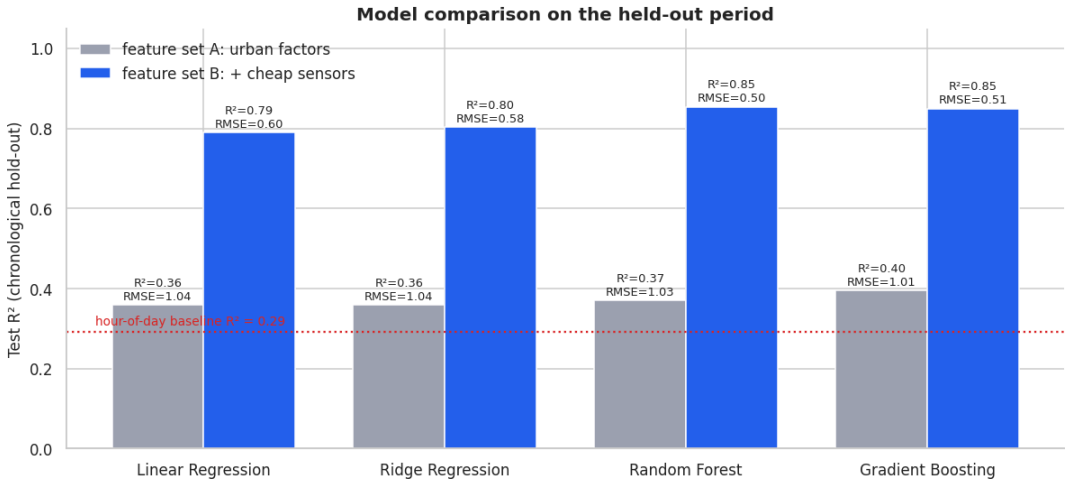


Figure 5. Test R^2 and RMSE for each model and feature set. The dotted line is the hour-of-day baseline.

Three results matter.

- Time and weather alone explain only about a third of the variation. With Set A, the models scored R^2 between 0.36 and 0.40, and the hour-of-day baseline alone already reached 0.29. The remaining two thirds is driven by things not in the data, such as traffic counts, wind, and the weather that decides whether pollution disperses.
- Adding the five cheap sensors lifted R^2 to 0.86 and halved the error, from about 1.03 to 0.49 mg/m³ with the tuned Random Forest. This is the case for low-cost sensors: a sensor box plus a model gets within half a milligram of a certified analyser.
- The models show the bias and variance trade-off. Linear Regression had training R^2 of 0.91 and test R^2 of 0.79 with Set B, so its scores stayed close together. The Random Forest fitted the training data almost perfectly (R^2 0.986) but scored 0.854 on unseen months. Tuning helped only slightly. The best settings were a depth of 12 and half the features tried at each split, and they moved test RMSE from 0.497 to 0.490.

We also repeated the Set A forest with a random split, to see what we avoided. The shuffled split reported R^2 of 0.616 instead of 0.370, an overstatement of 0.246, because near-identical neighbouring hours landed on both sides of the split. That is why we used a chronological split.

We saved the model and tested it on a real hour. The model file was loaded fresh from disk and given the inputs for 18:00 on Thursday 3 February 2005. It predicted 6.96 mg/m³ of CO, and the analyser measured 6.70. When we gave the urban factors model the same weather at 18:00 and then at 03:00, it predicted 3.19 and 0.60 mg/m³, which shows it responds to the hour of day.

What drives pollution

The linear coefficients are a numerical version of the hourly pattern. Relative to midnight, the hours from 17:00 to 20:00 add 1.4 to 2.1 mg/m³ of CO, 08:00 to 09:00 adds about 1.1, and 03:00 to 06:00 subtracts about 1. For the urban factors forest, hour of day dominated, because shuffling it raised test RMSE by 0.5 mg/m³, three to four times more than the next feature. Once the sensors were added, the S2 sensor carried most of the signal (shuffling it cost about 0.77 mg/m³), and the time and weather features barely mattered (Figure 6).

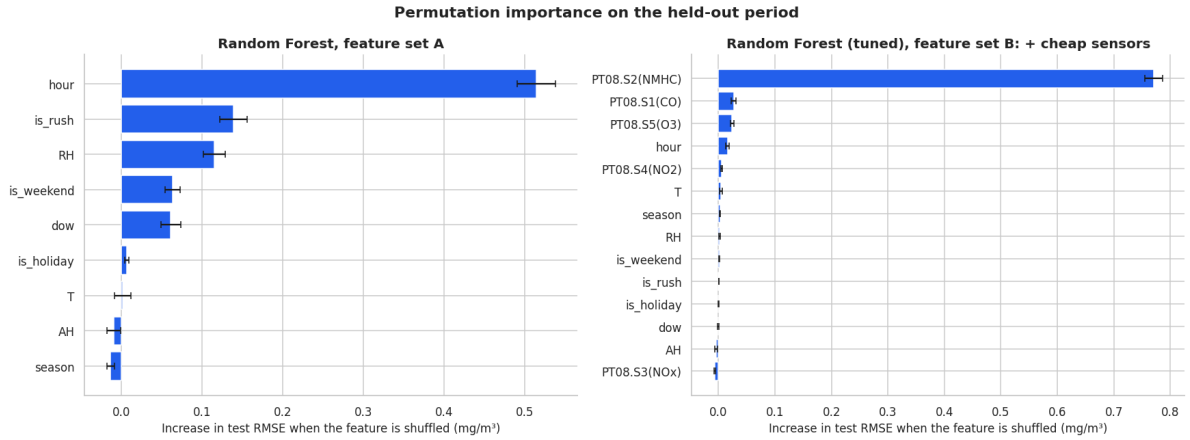


Figure 6. Permutation importance: how much test error rises when each input is shuffled.

Controlling for the hour, partial dependence plots showed what weather really does. Predicted CO is flat until relative humidity reaches about 40%, then rises by about 0.7 mg/m³ towards 80%, which fits still, damp evenings when air stops mixing. Temperature has a small negative effect of about 0.2 mg/m³ between 10 and 35 °C. So the raw weather correlations were near zero not because weather does nothing, but because its effect was hidden behind the time-of-day cycle. For the brief's rainfall question, the closest honest answer is that high humidity goes with more CO, not less.

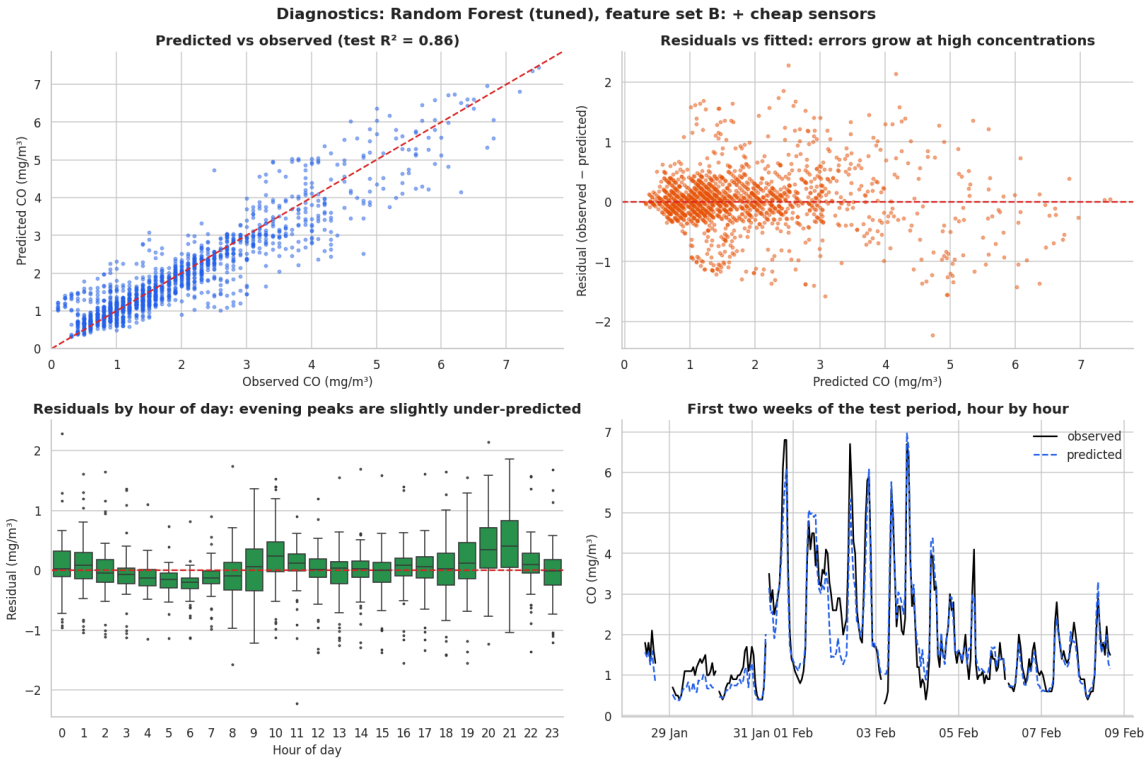


Figure 7. Diagnostics for the best model: predicted versus observed, residuals, and the first two weeks of the test period.

The model is weakest at peaks. It tracks observed values up to about 4 mg/m³ and then spreads out, and its biggest misses are under-predictions, by about 0.3 mg/m³ at 20:00 to 21:00. That is the wrong direction for an alert system, so a city should set its alert threshold lower to compensate.

6. Conclusion and Recommendations

The main insight is that when pollution happens matters more than what the weather is. We recommend four actions.

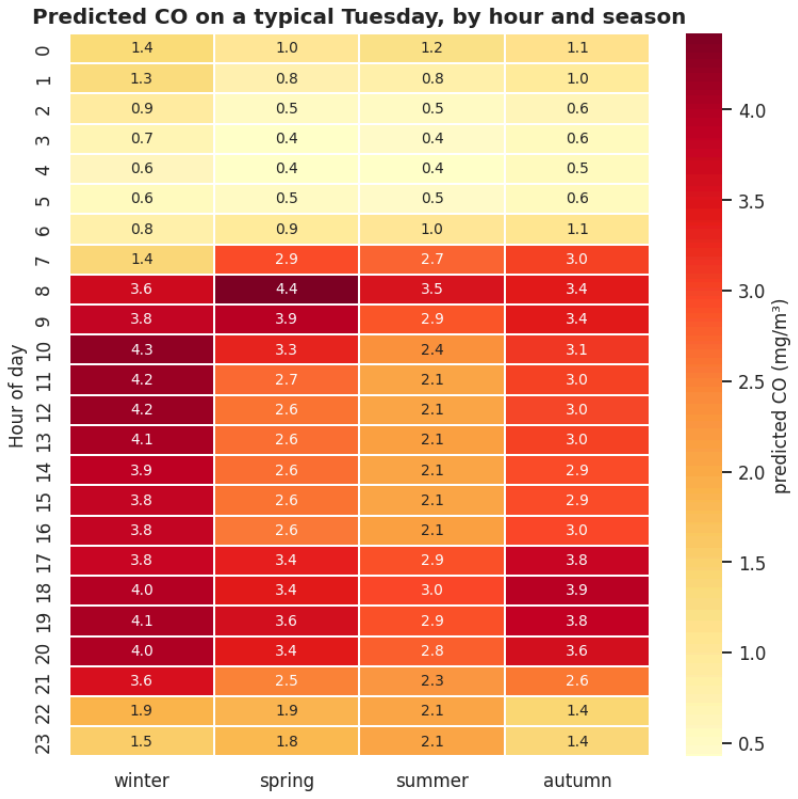


Figure 8. Predicted CO on a typical Tuesday by hour and season, from the urban factors model.

- Target weekday rush hours, not whole days. Hour of day is the strongest predictor in every model, and the harm is concentrated in about six weekday hours. A restriction or congestion charge in the windows 07:00 to 10:00 and 17:00 to 21:00 would address most breaches at a lower cost than an all-day measure. Figure 8 gives the exact hours for each season.
- Treat winter differently. From November to February the model predicts high CO across the whole working day, with 3.6 to 4.3 mg/m³ from 08:00 to 21:00 on a typical Tuesday (Figure 8). All 15 WHO breach days fell in those months. An all-day measure is justified in winter, and humidity above about 50% is a useful warning signal.
- Deploy low-cost sensors, but only with a reference station and a regular recalibration schedule. Otherwise, as the drift analysis shows, the network would under-report winter pollution.
- Report against the right standards. Reporting only the EU 8-hour CO limit would call this site clean, so also report the WHO guideline and the NO2 hourly limit.

Limitations

- One site and one year. The hourly and weekly shapes should apply to other traffic-dominated sites, but the exact levels will not, and one year cannot separate a normal winter from an unusual one.
- No particulate matter, rainfall, wind or traffic counts. We infer that traffic is the source from the rush-hour shape, but never observe it. Wind is probably the biggest missing factor, and it is the likely reason two thirds of the variation is unexplained.
- Sensor data is a proxy for air quality, and we saw it drift. The reference analysers were also offline for 18% of hours, so some months carry less weight than others.
- The results show association, not cause. We cannot say that cutting traffic at 19:00 would lower CO by a set amount. That needs data from before and after a real policy change.

A decision-ready study would need several sites, at least three years of data, wind and rainfall records, traffic counts and a particulate matter analyser.

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